Math 141, S10 Name: Answer

Quiz 11 Section: SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (4 points) Use Part I of the Fundamental Theorem of the Calculus to find the derivatives of the following functions:

(a) $g(x) = \int_1^x \frac{1}{t^3+1}\,dt$

$$g'(x) = \frac{1}{x^3+1}$$

(b) $h(x) = \int_1^{x^2} t \sin t\,dt$

Let $u = x^2$

$$h'(x) = \frac{d}{du}\left[\int_1^u t\sin t\,dt\right]\frac{d}{dx}[x^2]$$

$$= (u\sin u)\cdot(2x)$$

$$= (x^2\sin x^2)\cdot(2x)$$

$$= 2x^3\sin x^2$$

Problem 2. (6 points) Evaluate the following definite integrals:

(a) $\int_0^2 (x^2-2x)\,dx$

$$= \left[\frac{x^3}{3} - x^2\right]_0^2$$

$$= \left[\frac{2^3}{3} - 2^2\right] - [0]$$

$$= -\frac{4}{3}$$

(b) $\int_{\pi/2}^{\pi} \cos\theta\,d\theta$

$$= [\sin\theta]_{\pi/2}^{\pi}$$

$$= \sin\pi - \sin\frac{\pi}{2}$$

$$= 0 - 1 = -1$$